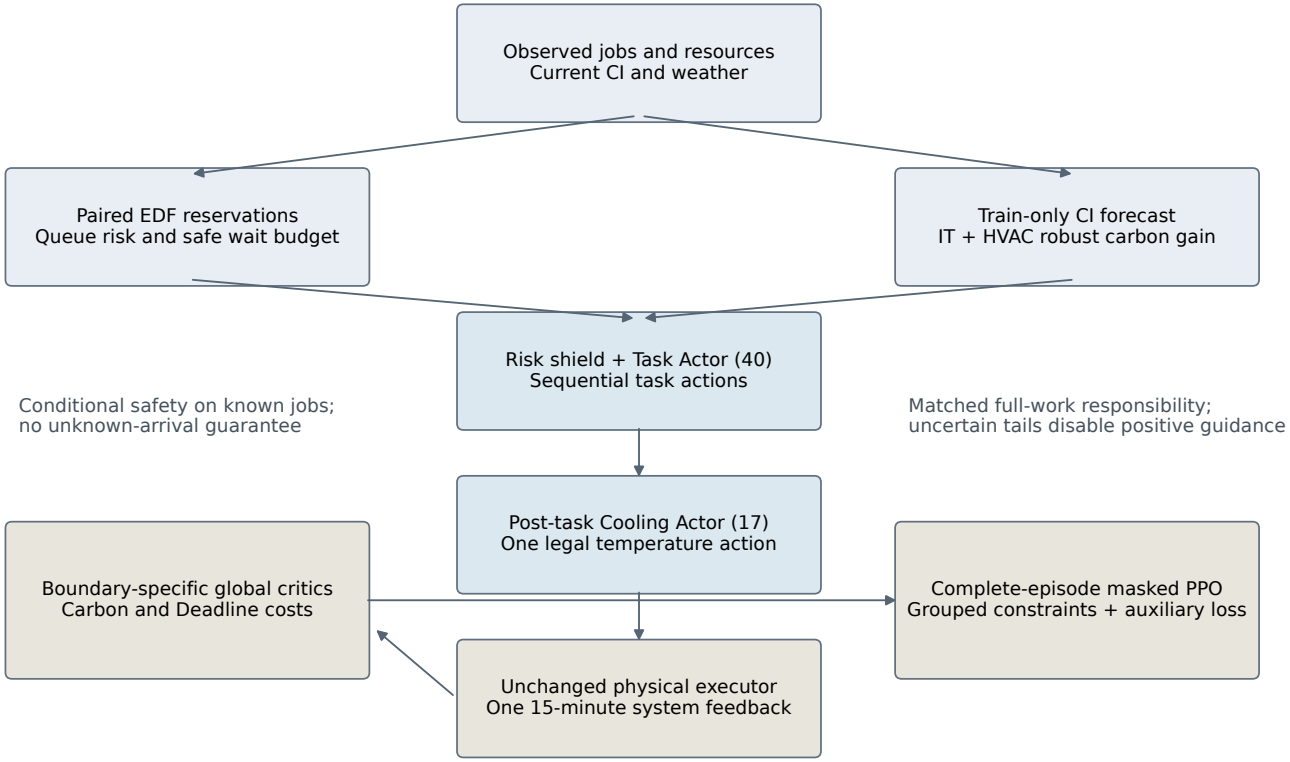
RAC-MAPPO: causal risk and carbon guidance
Observed jobs and resources
Current CI and weather
Paired EDF reservations
Queue risk and safe wait budget
Train-only CI forecast
IT + HVAC robust carbon gain
Risk shield + Task Actor (40)
Sequential task actions
Conditional safety on known jobs;
no unknown-arrival guarantee
Matched full-work responsibility;
uncertain tails disable positive guidance
Post-task Cooling Actor (17)
One legal temperature action
Boundary-specific global critics
Carbon and Deadline costs
Complete-episode masked PPO
Grouped constraints + auxiliary loss
Unchanged physical executor
One 15-minute system feedback